Exploratory Data Analysis (EDA) Summary Report

Project: Geldium Credit Delinquency Risk Assessment

Prepared for: Tata iQ Analytics & Geldium Decision-Makers

Status: Baseline Evaluation Phase

1. Introduction

The purpose of this report is to deliver a comprehensive exploratory data analysis (EDA) of Geldium's customer account dataset. This initial structural evaluation aims to audit dataset completeness, identify underlying historical risk trends, and pinpoint formatting inconsistencies. Establishing this high-quality, verified data foundation is a critical prerequisite before deploying any predictive AI/ML models designed to flag at-risk accounts and optimize collections intervention strategies.

2. Dataset Overview

A structural assessment matching the Dataset Description Guide against the raw source file reveals the current composition of the data array:

Key Dataset Attributes

- Number of Records: The production dataset comprises a comprehensive ledger of unique customer portfolios tracking sequentially from CUST0001 onward.
- Key Variables: Contains 19 core analytical features covering customer demographics, fundamental financial metrics (income, balances, debt ratios), and a rolling 6-month behavioral payment tracking history (Month_1 to Month_6).
- Data Types:
    - Numerical Fields: Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, and Account_Tenure.
    - Categorical Fields: Customer_ID, Employment_Status, Credit_Card_Type, Location, and the 6-month historical tracking variables.
    - Binary Target: Delinquent_Account ($0 =$ No, $1 =$ Yes).

Critical Data Mismatch Identified

- Schema Drift Exception: The *Dataset Description Guide* states that fields Month_1 through Month_6 should be stored as categorical integers ($0 =$ On-time, $1 =$ Late, $2 =$ Missed). However, the actual dataset contains explicit text strings ("On-time", "Late", "Missed"). This requires mapping back to numerical matrices during tokenization or feature engineering.

3. Missing Data Analysis

Maintaining model accuracy requires a structured resolution for unrecorded metrics. A line-by-line audit highlights targeted column data gaps:

Key Missing Data Findings

- Variables with Missing Values: The field Loan_Balance contains clear structural gaps, notably missing values for records such as CUST0009 (Row 10), CUST0024 (Row 25), and CUST0029 (Row 30). Additionally, the Income parameter exhibits sparse gaps across broader target groups.

Proposed Data Quality Treatment Plan

Feature Name	Identified Issue	Selected Handling Method	Technical Justification
Loan_Balance	Gaps in active balances (CUST0009, CUST0024).	Zero Imputation (0)	Gaps likely indicate the customer holds no active loan liabilities rather than unrecorded omissions.
Income	Hidden gaps across newer client accounts.	Median Imputation	Protects the baseline dataset from skew caused by high-income outliers.
Employment_Status	Truncated strings and structural inconsistencies.	Categorical Mapping & Cleaning	Standardizes variation arrays to prevent machine learning model confusion.

4. Key Findings and Risk Indicators

Uncovering interactions between customer behavior and account defaults reveals strong predictors of financial risk:

Key Findings & Visual Correlations

- **Utilization vs. Delinquency:** Accounts displaying elevated Credit_Utilization profiles alongside sustained "Late" or "Missed" sequences in the Month_1 to Month_6 tracking matrices show a strong correlation with a active Delinquent_Account status of 1 (e.g., CUST0002 and CUST0024).

- **Credit Score Erosion:** Lowered baseline credit scores (e.g., values under 500 such as CUST0001 with a score of 398) closely correlate with elevated trailing Missed_Payments counts.

Unexpected Structural Anomalies

- **Employment_Status Label Fragmentation:** The values in column J are highly un-standardized. There is a redundant mix of duplicated designations such as "EMP" vs. "Employed", truncated variables like "Self-employe", and non-standard capitalization choices like lowercase "retired" alongside capitalized variants. This string entropy will degrade categorical encoding models if left un-aligned.

5. AI & GenAI Usage

Generative AI frameworks were leveraged to accelerate core data analysis steps while maintaining strict data confidentiality boundaries. The analytical scaffolding used during the analysis phase relied on the following prompts:

- *“Summarize key patterns, structural anomalies, and text string mismatches between the provided schema definition document and the active production database rows.”*
- *“Suggest an industrial imputation strategy for missing consumer financial metrics based on banking domain best practices where extreme positive value outliers exist.”*

6. Conclusion & Next Steps

The foundational data shows clear predictive potential, but requires preprocessing to fix data quality issues before modeling can begin.

Recommended Next Steps

1. **Execute Data Type Realignment:** Convert the text labels ("On-time", "Late", "Missed") in Month_1 through Month_6 back to standard numerical categories (0,1,2) to match the official system documentation.
2. **Standardize Categorical Fields:** Run text-cleansing scripts across Employment_Status to consolidate fragmented labels like "EMP", "Employed", and truncated variants into clean, unified categories.
3. **Apply Imputation Rules:** Run the planned median and zero imputation routines across the verified Loan_Balance and Income gaps to ensure a complete, stable data matrix.